UAS DHARWAD SOIL SCIENCE & AGRONOMY DIAGNOSTIC LAB

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA07	**Report Date:**	21-Oct-2026
Location / District:	Navalgund Taluk, Dharwad District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Medium Black Soil (Vertic Inceptisol)	**Target Agro-Zone:**	Karnataka - Dharwad Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**280.0**	kg/ha	280.0 - 560.0	Medium (Sufficient)
Available Phosphorus	**P**	**41.5**	kg/ha	23.0 - 56.0	Medium (Adequate)
Available Potassium	**K**	**295.0**	kg/ha	140.0 - 330.0	Medium-High (Good)
Soil Reaction (pH)	**pH**	**7.50**	pH scale	6.5 - 7.5	Slightly Alkaline (Ideal)
Electrical Conductivity	**EC**	**0.56**	dS/m	< 1.00	Normal (Safe)
Organic Carbon	**OC**	**0.68**	%	0.50 - 0.75	Medium (Favorable)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 280.0 kg/ha [Medium (Sufficient)]

Available Phosphorus (P): 41.5 kg/ha [Medium (Adequate)]

Available Potassium (K): 295.0 kg/ha [Medium-High (Good)]

Soil Reaction (pH): 7.50 [Slightly Alkaline (Ideal)]

Electrical Conductivity (EC): 0.56 dS/m [Normal (Safe)]

Organic Carbon (OC): 0.68 % [Medium (Favorable)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: Productive medium black soil with high base saturation and good cation balance. Ideal soil buffer capacity and moisture retention under rainfed and semi-irrigated setups.

General Nutrient Management Guidance: Adopt Broad Bed and Furrow (BBF) conservation farming. Apply basal Zinc Sulphate @ 25 kg/ha every 2 years to counter micro-nutrient immobility in alkaline soil. Highly recommended for cotton, maize, chickpea, and sorghum.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*